

TITLE PAGE

- Problem Statement ID – SIH26174
- Problem Statement Title - AI Human Activity Recognition for On-board BAS Experiments
- Theme - Space Technology
- PS Category- Software
- Team ID- 162833
- Team Name - Team Samay



! KEY CHALLENGES

1

Zero Cloud / Network Unavailability

HIGH

No reliable ground link exists for real-time inference, guidance or data lookup during experiments.

- Cloud AI and remote assistance not usable
- Continuous autonomous operation required

2

Microgravity Orientation Shifts

HIGH

Astronauts and tools float freely, so body posture and "up/down" cues are unreliable.

- Vision models trained on gravity-based poses degrade
- Absolute vertical axis cannot be assumed

3

High Compute on Limited Power

MEDIUM

Multiple perception and reasoning models must run in real time on a tight power and thermal budget.

- Multi-camera 1080p streams are compute heavy
- CPU / memory overhead must stay low

4

Astronaut Tool Occlusion

MEDIUM

Hands and body frequently block tools and small objects from camera view mid-action.

- Risk of missed or false step detection
- Single-frame decisions are unreliable

Impact: Missed steps | Wrong object usage | Experiment failure | Higher crew workload



✓ ARCHITECTURE SOLUTIONS

1

Fully Offline Edge Architecture

Solves: Zero cloud

End-to-end processing on NVIDIA Jetson AGX Orin with no ground-link dependency.

- Local quantized LLM (Llama-3-8B GGUF) with offline RAG
- Piper TTS for autonomous voice assistance

2

Coordinate Normalization Module

Solves: Microgravity shifts

Poses and object positions are expressed relative to the payload rack instead of gravity.

- Fixed payload rack anchors as reference frame
- No reliance on absolute vertical axes

3

Optimized Edge Inference

Solves: Limited power

Models and video path are tuned to fit the power envelope without losing real-time speed.

- TensorRT optimization and INT8 quantization for YOLO / Swin
- Zero-copy GStreamer pipelines cut CPU / memory load

4

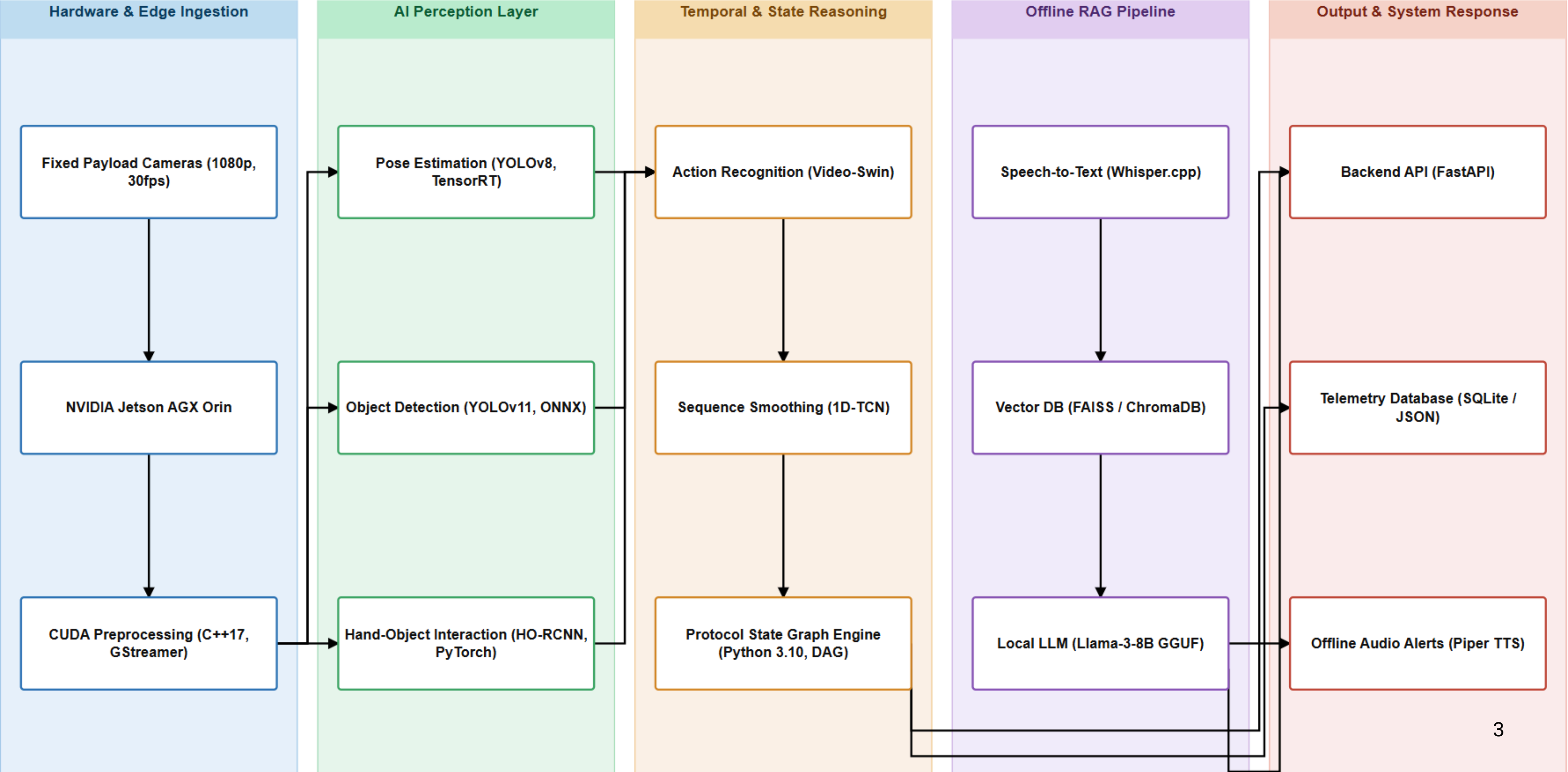
Temporal Reasoning & Smoothing

Solves: Tool occlusion

Actions are inferred across frames, so brief occlusions do not break step detection.

- 1D-TCN sequence smoothing filters false positives
- Multi-frame reasoning infers actions when tools are hidden

Outcome: Reliable offline operation | Robust in microgravity | Power-efficient | Fewer false alerts





Technical Feasibility

- **Nothing here is experimental.** YOLO and Video-Swin are well tested, and we only have to fine-tune pretrained weights on lab-procedure footage.
- **Enough horsepower on board.** Jetson AGX Orin gives up to 275 TOPS at a 15-60 W setting, so detection, pose and action models can run together in real time.
- **Low-latency video path.** Frames go from camera to GStreamer to CUDA without extra CPU copies, even with several 1080p feeds.
- **Voice and guidance stay local.** Whisper.cpp, a quantized Llama-3-8B and Piper TTS all run offline on the same unit.



Operational Readiness

- **Fits where ISRO is heading.** Longer missions and the Bharatiya Antariksh Station mean waiting on ground support won't always be an option.
- **No ground link needed.** Experiments keep going through communication delays or blackouts.
- **Built for floating crew.** Positions are measured against fixed rack anchors, not "up" and "down", so body orientation doesn't confuse the models.
- **Flexible by design.** Steps live in an editable state graph, so a different experiment can be loaded without retraining.

Bottom line: every building block already exists and runs on edge hardware. The real work is integration and fine-tuning, which is a realistic scope for a working prototype.



Economic Viability

- **Far less data sent to Earth.** Only short logs and reports are downlinked, not raw video, so satellite bandwidth use drops a lot.
- **Lighter training load.** Astronauts get step-by-step help during the experiment, so fewer hours go into pre-flight procedure drills and refreshers.
- **Mostly software.** A new experiment needs a new protocol file, not new hardware, which keeps replacement and upgrade costs low.
- **Easy to copy.** The same setup can be repeated on other payload racks by adding cameras and loading that rack's protocols.



Risk Mitigation

- **Human stays in charge.** When confidence is low, the system asks the astronaut to confirm instead of guessing. Manual confirmation always works.
- **Grounded answers.** The assistant pulls from a local vector database of the experiment's own approved documents.
- **Fewer false alarms.** Temporal smoothing looks across frames, so one blurry or blocked frame won't trigger a wrong alert.
- **Full paper trail.** Every decision is timestamped and logged, so ground teams can review exactly what happened.

Still to do, honestly: collect and label enough procedure footage for fine-tuning, and later test the unit for flight conditions like radiation, vibration and thermal limits.



Impact on Target Audience

The solution supports multiple stakeholders in the space ecosystem.



Astronauts

- Real-time guidance & voice alerts
- Reduced cognitive load
- Support during complex procedures



Scientists & Researchers

- Reliable experiment execution
- Better data quality & traceability
- Structured experiment records



Mission Control / Ground Teams

- Lightweight status updates
- Less dependency on continuous raw video transmission



ISRO & Space Program Stakeholders

- Supports BAS and future missions
- Enables autonomous onboard operations



Benefits of the Solution

Creates value across social, economic and environmental dimensions.



Social Benefits

- Safer and more efficient space missions
- Supports human spaceflight and scientific progress
- Builds confidence in India's space capabilities



Economic Benefits

- Reduces mission risk and operational cost
- Optimizes use of onboard resources
- Enables scalable solution for multiple experiments



Environmental Benefits

- Minimizes unnecessary data transmission
- Lowers energy and bandwidth consumption
- Supports sustainable long-duration missions

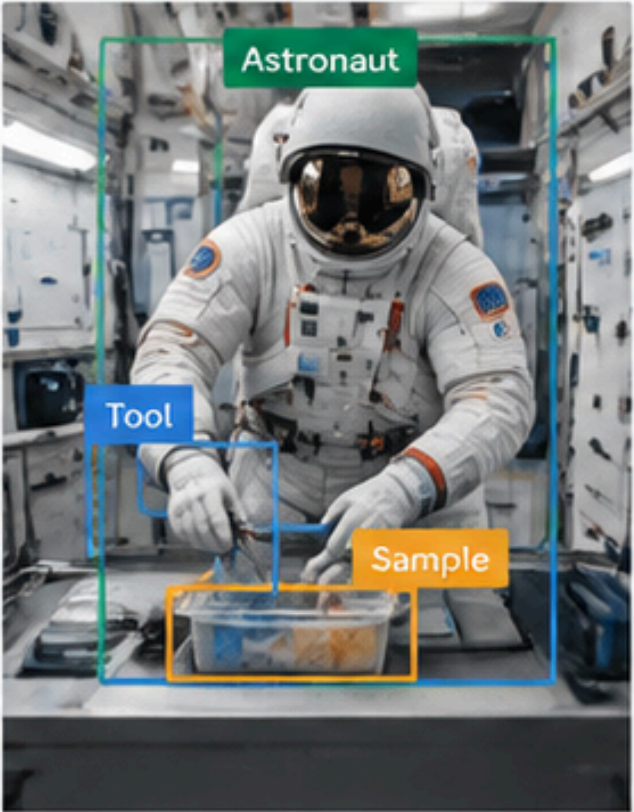


Scientific & Technological Benefits

- Improves experiment reliability and data quality
- Advances AI, computer vision and space technology
- Creates a foundation for future autonomous systems



Example: Detected Deviation



Expected Sequence

1. Open Container ✓
2. Take Sample ✓
3. Transfer Sample ✓
4. Close Container ✓

Observed Sequence

1. Open Container ✓
2. Take Sample ✓
4. Close Container ✗

Step 3 Skipped

Voice Alert Issued



Result

- ✓ Timestamped Experiment Log
- ✓ Local Video Storage
- ✓ Lightweight Status Report



Path to Long-Term Impact

From experiment assistance today to autonomous space missions tomorrow.



Earth-Based
Prototype



BAS Experiments
(Extended Testing)



On-board Assistance
(Operational Use)



Long-Duration
Missions



Future Deep-Space
Exploration

1 Vision & Temporal Action Recognition (AI Perception)



Video-Swin Transformer

- Models complex temporal sequences and recognizes astronaut actions in microgravity environments.
- Reference: Liu, Z., et al. "Video Swin Transformer."
<https://arxiv.org/abs/2106.13230>



YOLOv8 & YOLOv11 (Ultralytics)

- High-speed, low-latency tracking of space payload tools (pipettes, vials) and real-time human pose estimation.
- Reference: "What is YOLOv8: An In-Depth Exploration..."
<https://arxiv.org/abs/2408.15857>



Hand-Object Interaction (HO-RCNN)

- Validates exact tool-contact states (holding, operating) in zero-gravity environments.
- Reference: "Hand-Object Interaction Detection With Fully Convolutional Networks" (CVPR).
<https://openaccess.thecvf.com/>

2 Sequence Smoothing & Protocol Logic



1D-Temporal Convolutional Networks (1D-TCN)

- Handles temporary visual occlusions (e.g., tools blocked by hands) and filters false-positive actions.
- Reference: Lea, C., et al. "Temporal Convolutional Networks for Action Segmentation and Detection."
<https://arxiv.org/abs/1611.05267>

3 Zero-Cloud Offline RAG Pipeline (Cognitive AI)



Llama-3-8B-Instruct (Quantized GGUF)

- Instruction-tuned LLM for offline edge inference (NVIDIA Jetson).
- Ensures autonomous operation without Earth-link dependencies.
- Reference: Meta AI. "The Llama 3 Herd of Models."
<https://arxiv.org/abs/2407.21783>



FAISS (Vector Database)

- Indexing and querying official ISRO BAS experiment manuals in a localized, fully offline vector space.
- Reference: Johnson, J., et al. "Billion-scale similarity search with GPUs."
<https://arxiv.org/abs/1702.08734>

4 Space-Grade Edge Computing & ISRO Context



Gaganyaan Mission Alignment

- Supports ISRO's Gaganyaan objective of Onboard Autonomous Procedure Monitoring (OAPM).
- Enhances crew safety and reduces reliance on ground control.
- Reference: ISRO Official Gaganyaan Mission Parameters.
<https://www.isro.gov.in/Gaganyaan.html>



Edge AI Feasibility in Space

- Uses localized NVIDIA compute nodes with CUDA-accelerated GStreamer zero-copy pipelines.
- Bypasses telemetry bandwidth limits and communication latency.

5 Additional References & Tools



Python
(Development)



VS Code
(IDE)



NVIDIA Jetson
(Edge Compute)



GStreamer
(Video Pipeline)



ISRO BAS Manuals
(Domain Knowledge)